Report

Of

Internship

Submitted

To

School of Computer Science and Engineering IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

MCA



By

Roll Number of Student: 25 SCS2040003499

Name of Student: Ayush Kumar

Batch: 2025- 2027

2026- 27

1. CANDIDATE'S DECLARATION

I, Ayush Kumar do hereby declare that the internship report titled "Java Software Development Projects" has been completed by me to fulfil the requirement for the award of the degree of MCA . The report presents the work completed during the internship, including the Student Grade Tracker, Stock Trading Platform, Artificial Intelligence Chatbot, and Hotel Reservation System projects. All references used in preparing this report have been acknowledged appropriately. I affirm that this report is my own work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____
Student Name: Ayush Kumar
Roll No.: 25SCS2040003499

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, constructive criticism, and friendly advice during this work. I am sincerely grateful to the internship organization, CodeAlpha, for providing an opportunity to apply programming concepts through practical project development.

I would also like to express my gratitude to the faculty members of the School of Computer Science and Engineering, IILM University, Greater Noida, for their academic guidance and support. Their encouragement helped me understand software development concepts and organize the internship work in a systematic manner.

I express my gratitude to my parents for their blessings and continuous support throughout the internship and academic journey.

Signature: _____

Student Name: Ayush Kumar

Roll No.: 25SCS2040003499

3. INTERNSHIP COMPLETION CERTIFICATE

CodeAlpha Internship Offer Letter

Dear Ayush Kumar

Congratulations!

We are pleased to inform you that you have been successfully selected for the Java Programming Internship Program at CodeAlpha.

Your official Offer Letter has been successfully generated and attached with this email as a PDF document. Kindly review the document carefully for internship details, duration, guidelines, and responsibilities.

Internship Details
Intern Name: Ayush Kumar

Profile: Java Programming

Internship Duration: 10st Sept 2026 - 10th October 2026

Offer Letter Issue Date: 2nd September 2026

CODE ALPHA

YOUR DREAM OUR PASSION

INTERNSHIP OFFER LETTER

Date: 2nd September 2026 Student ID: CA/DF1/268905

Name: Ayush Kumar

Welcome to CodeAlpha.

Congratulations! We are happy to offer you the position of **Java Programming** Intern at CodeAlpha. This letter confirms your selection for our Virtual Java Programming Internship, starting from **10th September 2026 to 10th October 2026**.

The internship will be conducted completely online and will give you an opportunity to work on practical assignments and projects related to Java Programming. You will be able to apply what you already know, improve your technical skills and gain hands-on experience.

Throughout the internship, you are expected to participate actively, complete the assigned tasks on time and follow the guidelines shared by CodeAlpha. We also expect you to maintain professional conduct and communicate responsibly during the internship.

A performance-based stipend may be provided depending on your performance, participation, contribution and completion of assigned work, as per the applicable terms of the internship program.

We are pleased to have you join us and look forward to your participation. We hope this internship will be a valuable and enjoyable experience for you.

We wish you all the best for your internship with CodeAlpha.

Best regards,

Swati Srivastava
Co-Founder & CEO

CODEALPHA SOFTWARE DEVELOPMENT

सत्यमेव जयते

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4. PROJECT DESCRIPTION

During the internship, four Java-based software projects were developed to demonstrate practical application of programming, Object-Oriented Programming (OOP), data management, user interaction, problem-solving, and persistence concepts. The completed projects are: (1) Student Grade Tracker, (2) Stock Trading Platform, (3) Artificial Intelligence Chatbot, and (4) Hotel Reservation System.

4.1 Introduction

During the internship, I gained hands-on experience in Java programming and software development. My work centered on transforming problem statements into working applications by identifying key entities, designing classes and methods, handling user input, validating data, processing information, and delivering meaningful results.

Student Grade Tracker → A system that records student names and grades while generating statistical insights.

Stock Trading Platform → A simplified trading environment that incorporates market data, enables buying and selling, manages portfolios, and maintains transaction history.

Artificial Intelligence Project → Focused on applying AI concepts to practical scenarios, enhancing problem- solving and automation.

4.2 Organization Profile

CodeAlpha served as the internship organization for the project work. The internship assignments were designed around practical software development activities, enabling participants to build programming projects and showcase technical proficiency through completed applications.

For the final report submission, details such as the official organization profile, internship offer or selection communication, and any formal descriptions provided by the organization should be included from the official internship documents, if required by the college.

4.3 Problem Statement

The internship tasks were designed to solve practical application problems using Java. The first problem was to create a system for entering and managing student grades

and calculating useful statistics. The second problem was to simulate a stock trading environment where users could view market information, buy and sell stocks, and monitor their portfolio. The third problem was to develop an interactive chatbot capable of processing user messages and returning suitable responses using NLP-inspired rule-based logic. The fourth problem was to create a hotel reservation system that could search available rooms, manage bookings and cancellations, simulate payment, and persist booking and availability information using File I/O. The main challenge across the projects was to design understandable, maintainable, and user-friendly applications while applying OOP principles and suitable data structures.

4.4 Project Objectives

Enhance practical skills in Java programming and problem- solving.

Apply Object-Oriented Programming principles, including classes, objects, encapsulation, and methods.

Organize and manage application data using ArrayList and object-oriented structures.

Build interactive user interfaces to support real- time interaction.

Implement robust input validation and generate meaningful system feedback.

Perform calculations, manage data, and present information in a clear, structured format.

Gain exposure to basic Natural Language Processing techniques and rulebased chatbot logic.

4.5 Scope of the Project

The scope of the internship work is educational and demonstrates the implementation of core Java application concepts. The Student Grade Tracker can be extended with student IDs, subjects, letter grades, class-wise statistics, and persistent storage. The Stock Trading Platform can be extended with real-time market APIs, authentication, database storage, order types, and advanced portfolio analytics. The Artificial Intelligence Chatbot can be extended with larger datasets, external NLP libraries, machine-learning models, database-backed FAQs, and web deployment. The Hotel Reservation System can be extended with check-in/check-out dates, multiple-room bookings, customer authentication, online payment gateways, and a database such as MySQL.

The present implementations are simulations and academic projects. The stock trading application does not execute real financial trades, and the chatbot uses rule- based logic rather than a production machine- learning model.

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Java	Primary programming language
OOP	Designing classes, objects, and application logic
ArrayList	Managing dynamic collections of students, stocks, transactions, and FAQ data
Java Swing	Graphical user interface for the chatbot
Scanner	Console- based user input
Java Time API	Date and time responses in the chatbot
File I/O	Persistent storage where required by the project design
JDK / Java Compiler	Compiling and executing Java applications
IDE / Text Editor	Writing, testing, and debugging source code

4.7 System Architecture

The applications follow a modular, object-oriented architecture. User input is received through the console or GUI, processed by application logic, stored in appropriate objects or collections, and then displayed as output.

Student Grade Tracker Architecture: The Student class stores a student name and grades. The main application accepts input, creates Student objects, stores them in an ArrayList, and calls methods for average, highest, and lowest scores before generating a summary report.

Stock Trading Platform Architecture: The Stock class stores market information, Transaction stores buy/sell details, Portfolio manages holdings, User manages balance and trading actions, and the main platform controls market display and menu operations.

AI Chatbot Architecture: The GUI receives the user's message, preprocessing converts it to normalized text, the intent/FAQ logic compares keywords, a suitable response is selected, and the response is displayed in the chat window.

Project	Main Components	Data Flow
Student Grade Tracker	Student, Main Program, ArrayList	Input → Student Object → Grade Calculations → Report

Stock Trading Platform	Stock, User, Portfolio, Transaction	Market → User Action → Portfolio/Balance → Transaction History
AI Chatbot	GUI, FAQ Map, NLP Preprocessor, Response Logic	User Message → Preprocess → Match Intent → Response → GUI
Hotel Reservation System	Room, Reservation, Hotel	User → Search/Book → Reservation → File Storage

4.8 Methodology

The projects were developed using an incremental approach. First, the requirements of each task were identified. Next, the major entities and operations were converted into classes and methods. Suitable data structures were selected according to the nature of the information. Input validation was added to reduce invalid operations. File I/O was incorporated where persistent data was required. Finally, the applications were tested using representative inputs and expected outputs.

Requirement analysis – identify features and expected outputs.

System design – identify classes, attributes, methods, and relationships.

Implementation – write Java classes and application logic.

Data management – use ArrayList, objects, maps, and file storage where applicable.

Interface development – implement console menus and GUI interaction.

Testing – check valid input, invalid input, calculations, transactions, and responses.

Documentation – record the project purpose, implementation approach, outcomes, and future scope.

4.8.1 Project 1 – Student Grade Tracker

The Student Grade Tracker is a console-based Java application designed to input and manage student grades. Each student is represented as an object containing a name and an ArrayList of grades. The program accepts the number of students and subjects, validates grades within the 0–100 range, and calculates average, highest, and lowest scores.

Feature	Implementation
Student management	Student class and ArrayList<Student>

Grade storage	ArrayList<Double>
Average	Sum of grades divided by number of grades
Highest score	Iterative comparison of grade values
Lowest score	Iterative comparison of grade values
Report	Formatted console summary

4.8.2 Project 2 – Stock Trading Platform

The Stock Trading Platform is a simulated trading environment. It maintains a set of stocks with symbols, company names, and prices. A User object maintains cash balance, a Portfolio, and transaction history. Users can display market data, buy stocks, sell stocks, view holdings, view account value, view transactions, an simulate market price updates.

Feature	Implementation
Market data	Stock objects stored in an ArrayList
Buying	Checks balance and adds shares to portfolio
Selling	Checks holdings and returns cash
Portfolio	Stores stocks and quantities
Performance	Calculates current portfolio value
Transactions	Transaction objects record BUY/SELL operations
Price update	Simulated random market price movement

4.8.3 Project 3 – Artificial Intelligence Chatbot

The Artificial Intelligence Chatbot is a Java Swing desktop application for interactive communication. It uses a predefined FAQ knowledge base and rulebased response logic. User input is converted to lowercase, punctuation is removed, extra spaces are normalized, and words are compared with stored FAQ keywords. The system also includes simple date and time responses.

Feature	Implementation
GUI	Java Swing JFrame, JTextArea, JTextField and JButton
FAQ training	HashMap containing common questions and responses
NLP preprocessing	Lowercase conversion, punctuation removal and normalization
Intent matching	Keyword similarity scoring
Dynamic response	Date and time based responses

Fallback	Helpful message for unknown queries

4.8.4 Project 4 – Hotel Reservation System

The Hotel Reservation System is a Java console-based application for searching, booking, and managing hotel rooms. It categorizes rooms into Standard, Deluxe, and Suite types, provides simulated payment processing, allows users to view booking details and cancel reservations, and uses File I/O to store active booking information. When the application restarts, stored bookings are loaded and the corresponding rooms are marked as unavailable.

Key Features

Room categorization: Standard, Deluxe, and Suite rooms with different prices.
Room search: Displays available rooms according to the selected category.
Reservation: Creates a unique booking ID and reserves an available room.
Cancellation: Cancels an active reservation and restores room availability.
Payment simulation: Confirms payment before finalizing a booking. Booking details: Shows customer, room, amount, and payment status information.
Persistence: Stores active bookings in hotel_bookings.txt and reloads them at startup.
Implementation Method: The system initializes room inventory, loads previously saved reservations, accepts customer and room-category information, validates room availability, simulates payment, creates a reservation object, marks the room as booked, and saves the reservation to a text file. Cancellation reverses the room availability state and updates the saved booking data.

4.9 Expected Outcomes

The completed projects demonstrate practical understanding of Java programming and object-oriented application design. The Student Grade Tracker demonstrates data management and statistical calculations. The Stock Trading Platform demonstrates modeling of real-world entities and transactions. The AI Chatbot demonstrates GUI development, basic NLP processing, and rule-based artificial intelligence. The Hotel Reservation System demonstrates room inventory management, reservation workflows, simulated payment handling, and persistent storage using File I/O.

Improved understanding of Java syntax, classes, objects, methods, and collections.

Ability to translate a written problem statement into a working software solution.

Experience with console and GUI based interaction.

Understanding of data validation and application flow control.

Practical exposure to modular OOP design.

Basic understanding of NLP preprocessing and intent matching. Experience in

testing and documenting software projects.

4.10 Certificates of Completion and Other Communication Mail Proof

The internship completion certificate, offer/selection communication, and relevant communication e-mails or messages should be attached in this section as required by the university report instructions.

[ATTACH CERTIFICATE / EMAIL / COMMUNICATION PROOF HERE]

There is only offer letter email after the completion of this internship I'll submit the

Completion certificate.

CodeAlpha Internship Offer Letter Dear Ayush Kumar

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Intern Name: Ayush Kumar Profile: Java Programming

Internship Duration: 10st September 2026 – 10th October 2026
Offer Letter Issue Date: 2nd september 2026

5. BIBLIOGRAPHY / REFERENCES

Oracle Java Documentation – Java language and standard library concepts. Java Platform documentation for collections, Swing, file handling, and date/time APIs.

CodeAlpha internship task descriptions and project requirements provided to the participant.

IILM University, School of Computer Science and Engineering – Internship Report Format (2026–27).

APPENDIX – PROJECT SUBMISSION CHECKLIST

Project	Source Code	Testing	Report Coverage
Student Grade Tracker	Completed	Completed	Included
Stock Trading Platform	Completed	Completed	Included
Artificial Intelligence Chatbot	Completed	Completed	Included
Hotel Reservation System	Completed	Completed	Included

Note: Screenshots of source code, application execution, output, and testing may be inserted after the relevant project sections if required by the college or internship evaluator.